

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 3

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1. TITLE

Write a Program Using JavaScript (Arrays and Functions)

2. AIM

To write a JavaScript program using **arrays and functions** to perform basic array operations, calculate the sum and average, and find the largest number.

3. OBJECTIVE

- To understand arrays in JavaScript.
 - To perform basic array operations.
 - To understand JavaScript functions.
 - To use functions with and without parameters.
 - To calculate sum and average using an array.
 - To find the largest number using a function.
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4. THEORY

JavaScript is a scripting language used to create interactive and dynamic web applications. It provides various features such as **arrays, functions, loops, and methods** for processing data.

An **array** is a collection of multiple values stored in a single variable. Array elements are accessed using an index, starting from 0.

Common array operations include:

- `push()` – adds an element at the end of an array.
- `pop()` – removes the last element.
- `unshift()` – adds an element at the beginning.
- `shift()` – removes the first element.
- `length` – returns the number of elements in an array.

A **function** is a reusable block of code designed to perform a specific task. Functions can have parameters and can return values using the `return` statement.

In this program, arrays are used to store data, array methods are used to modify the data, and functions are used to perform operations such as **addition, finding the largest number, calculating sum, and calculating average**.

5. ALGORITHM / PROCEDURE

1. Create a JavaScript file.
 2. Declare arrays with different values.
 3. Display array elements using `console.log()`.
 4. Modify an array element using its index.
 5. Perform `push()`, `pop()`, `unshift()`, and `shift()` operations.
 6. Find the length of an array.
 7. Calculate the sum of array elements using a loop.
 8. Define a function without parameters.
 9. Define functions with parameters and return values.
 10. Create a function to find the largest number.
 11. Create a function to calculate the sum and average of an array.
 12. Execute the program using `Node.js`.
 13. Verify the output in the console.
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6. PROGRAM

Arrays.js

```
let colors = ["Red", "Green", "Blue"]; console.log(colors);
```

```
var fruits = ["Apple", "Banana", "Mango"];  
console.log(fruits[0]); console.log(fruits[1]);  
console.log(fruits[2]);
```

```
var fruits = ["Apple", "Banana", "Mango"];  
fruits[1] = "Orange"; console.log(fruits);
```

```
let marks = [85, 90, 78, 95]; console.log("marks:",marks);
```

```
//array methods let numbers =  
[10, 20, 30];  
console.log(numbers);  
console.log("after pushing 40");  
numbers.push(40);  
console.log(numbers);  
console.log("after popping");  
numbers.pop();  
console.log(numbers);  
console.log("after unshifting");  
numbers.unshift(5);  
console.log(numbers);  
console.log("after shifting");  
numbers.shift();  
console.log(numbers);
```

```
let numb = [10, 20, 30, 40]; console.log("array  
length:",numb.length);
```

```
let number = [10, 20, 30, 40]; let  
sum = 0; for (let i = 0; i <  
number.length; i++) {    sum +=  
number[i];  
}  
console.log("Sum =", sum);
```

```
//without parameters function greet(){
console.log("Welcome to JavaScript");
}
greet();
```

```
//with parameters function
add(a, b) {
console.log(a + b);
}
add(10, 20);
```

```
// Function to add two numbers
function add(a, b) { return a
+ b;
}
// Function call let result =
add(10, 20);
console.log("Sum =", result);
```

```
// finding largest number
function largest(a,b){
if (a>b){ return a;
} else { return b;
}
}
console.log(largest(15,25));
```

```
// Function to calculate sum and average
function calculateSum(arr) { let sum =
0; for (let i = 0; i < arr.length; i++) {
sum += arr[i];
}
return sum;
}
let num = [10, 20, 30, 40, 50]; let
total = calculateSum(num); let
average = total / num.length;
console.log("Array:", num);
```

```
console.log("Sum =", total);  
console.log("Average =", average);
```

7. CODE EXPLANATION

Element	discription
let colors = [...]	Creates an array.
fruits[0]	Accesses the first array element.
fruits[1] = "Orange"	Changes an array element.
push()	Adds an element at the end.
pop()	Removes the last element.
unshift()	Adds an element at the beginning.
shift()	Removes the first element.
length	Finds the number of elements.
for loop	Traverses the array.
function	Defines a reusable block of code.
return	Returns a value from a function.
console.log()	Displays the result in the console.

8. TEST CASES AND EXECUTION DATA

Test Case	Input / Action	Expected Result	Actual Result	Status
TC-01	Execute the JavaScript program	Arrays, array operations, sum, average, and largest number should be displayed correctly	Output displayed correctly	Pass
TC-02	numbers = [10,20,30], perform push(40)	Array should become [10,20,30,40]	Array updated correctly	Pass
TC-03	largest(15,25)	Largest number should be 25	25 displayed	Pass
TC-04	[10,20,30,40,50]	Sum = 150, Average = 30	Sum = 150, Average = 30	Pass

9. OUTPUT

Output :

```
Terminal
PS C:\Users\PAVAN\Desktop\fs\WEEKS\WEEK3> node arrays.js
[ 'Red', 'Green', 'Blue' ]
Apple
Banana
Mango
[ 'Apple', 'Orange', 'Mango' ]
marks: [ 85, 90, 78, 95 ]
[ 10, 20, 30 ]
after pushing 40
[ 10, 20, 30, 40 ]
after popping
[ 10, 20, 30 ]
after unshifting
[ 5, 10, 20, 30 ]
after shifting
[ 10, 20, 30 ]
array length: 4
Sum = 100
Welcome to JavaScript
Sum = 30
25
Array: [ 10, 20, 30, 40, 50 ]
Sum = 150
Average = 30
PS C:\Users\PAVAN\Desktop\fs\WEEKS\WEEK3>
```

10. RESULT

Thus, a JavaScript program using **arrays and functions** was successfully implemented. Basic array operations were performed, and the sum, average, and largest number were calculated successfully.
